

Discovering, Analyzing, and Disrupting Wildlife Trafficking Networks

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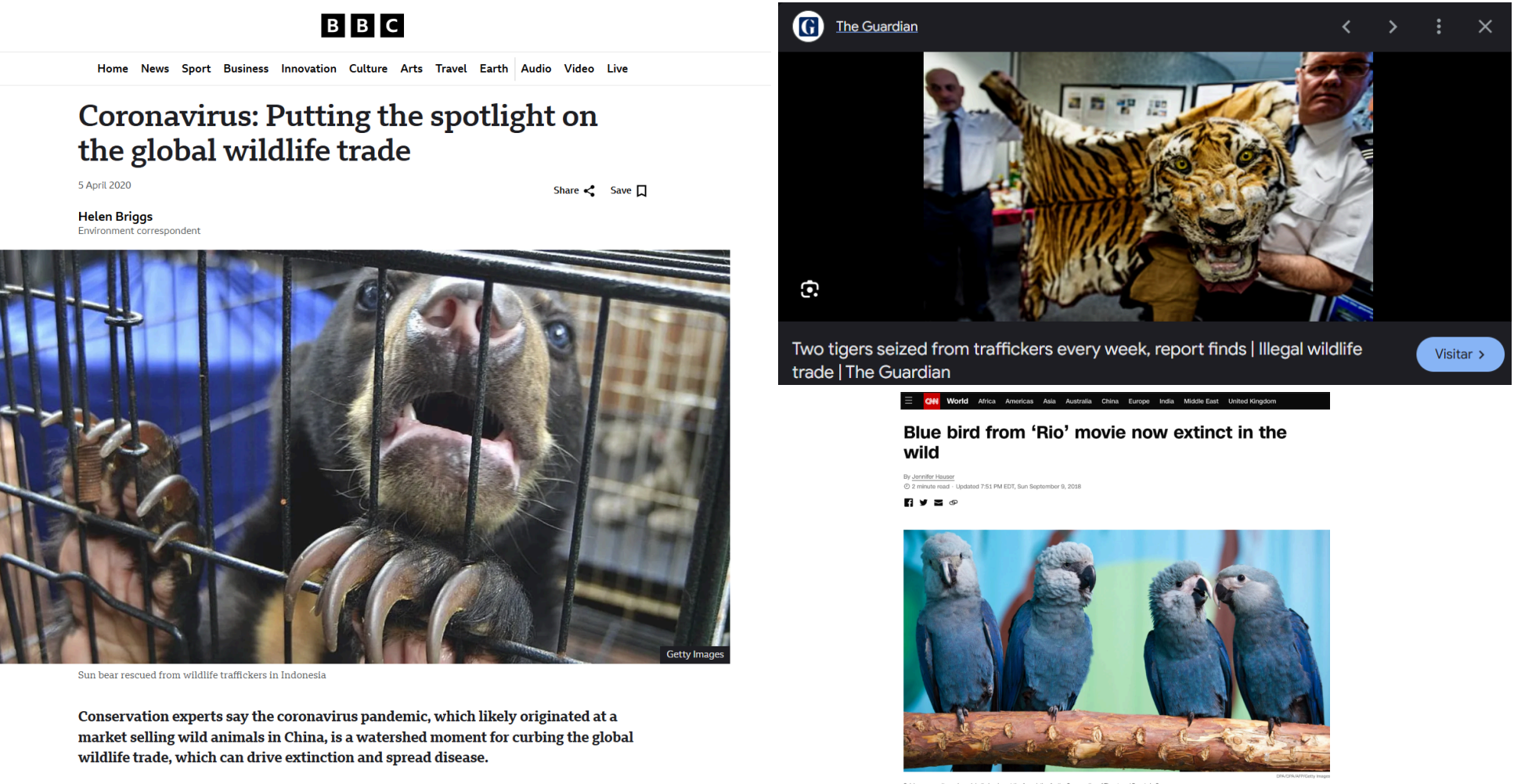


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Introduction

Wildlife trafficking is a critical global issue, significantly impacting biodiversity, ecological stability, and public health.



Illegal trade is rampant on open web-based platforms

Opportunity to discover digital traces (e.g., trafficking networks, patterns, trade volume) that enable a better understanding of trade and the design interventions to disrupt it.

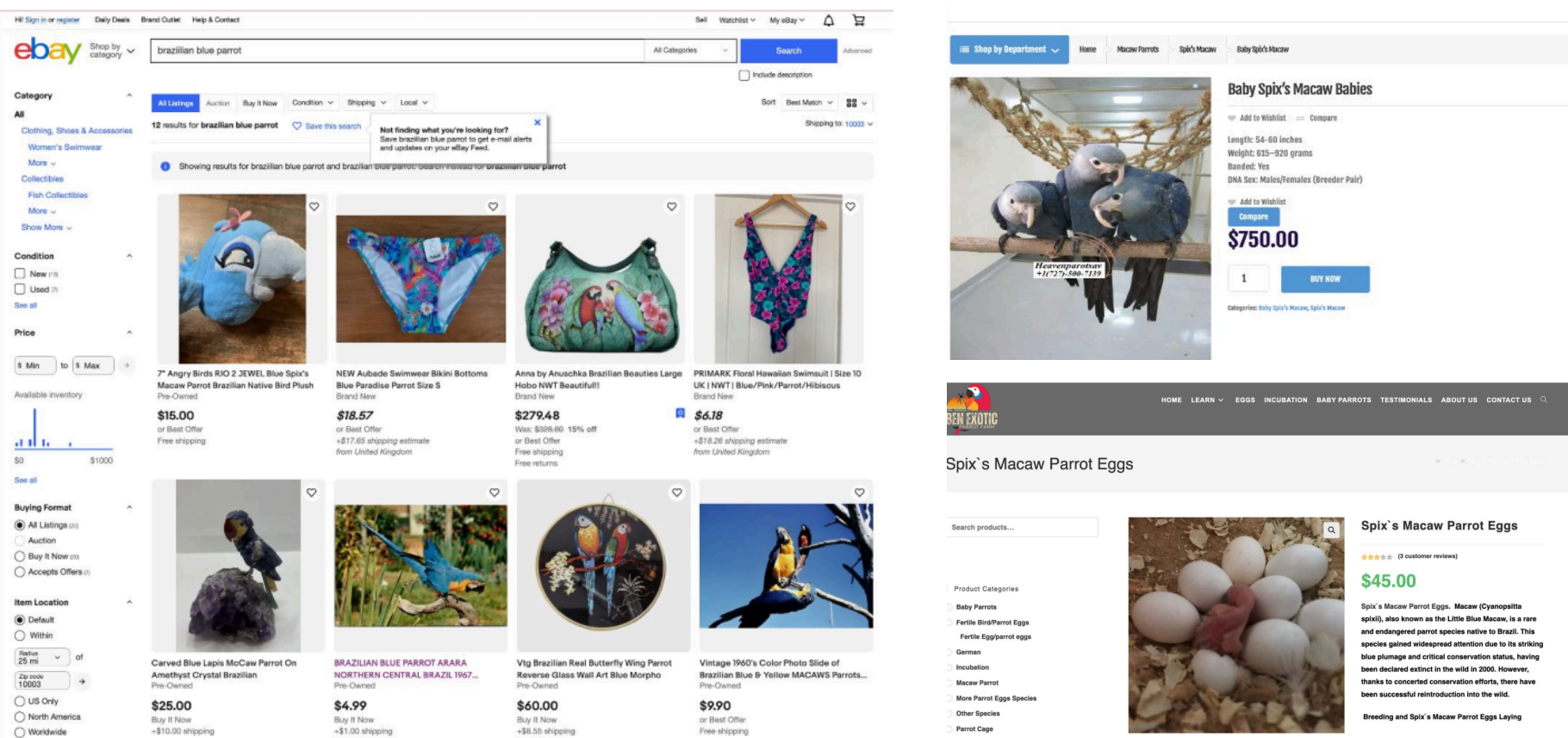
The Global Trade in Wild Animals and Plants

>35,000 terrestrial species are sold

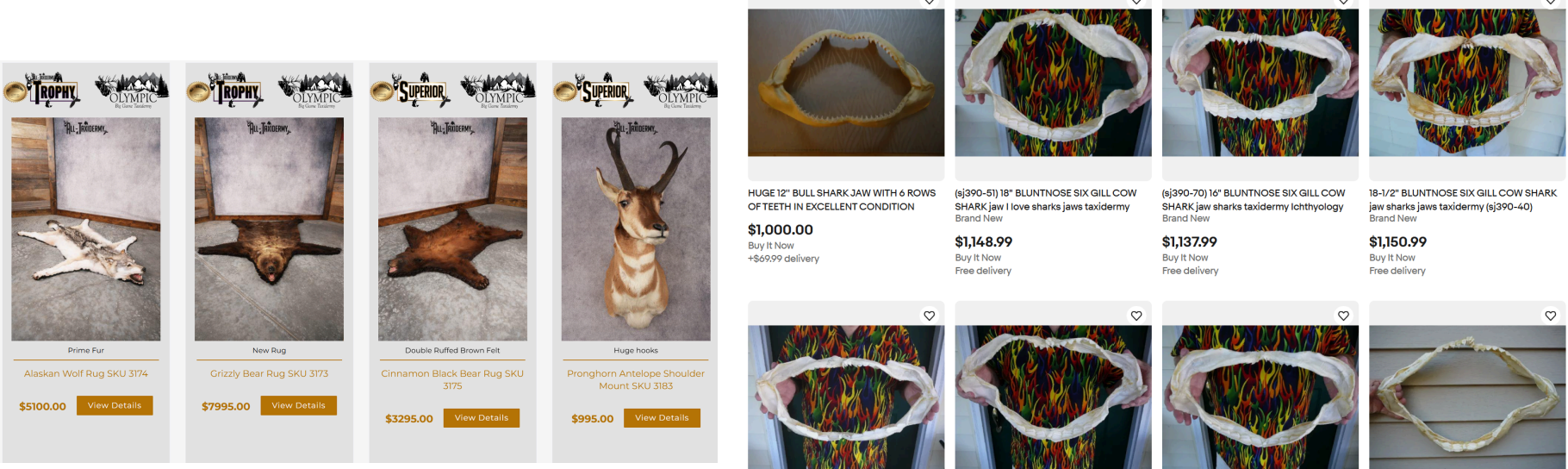
~\$300 billion *all* trade ~\$23 billion illegal trade (terrestrial)

~\$50-150 billion timber ~\$36 billion illegal (marine)

Challenge 1: Data Triage – Identify the few relevant wildlife ads among vast numbers of online listings – need to find a "needle in a haystack"



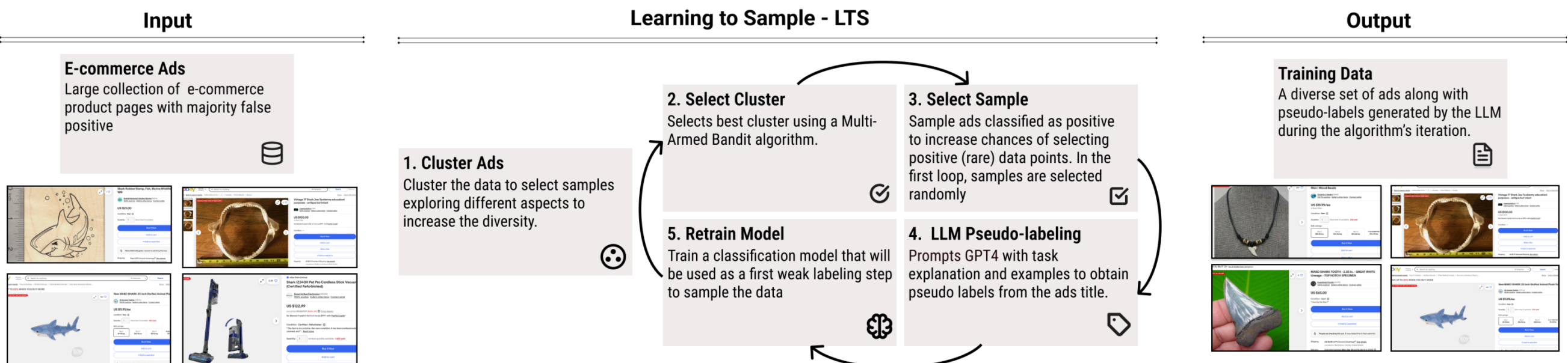
Challenge 2: Difficult to create learning-based classifiers – it is costly and time-consuming to label data



Using manual human labeling is expensive. If we use models as GPT-4 the cost is reduced, but it is still expensive. For example, to label 800,000 ads collected over one week would cost over \$17,000.

Our Approach

LTS - Learn To Sample

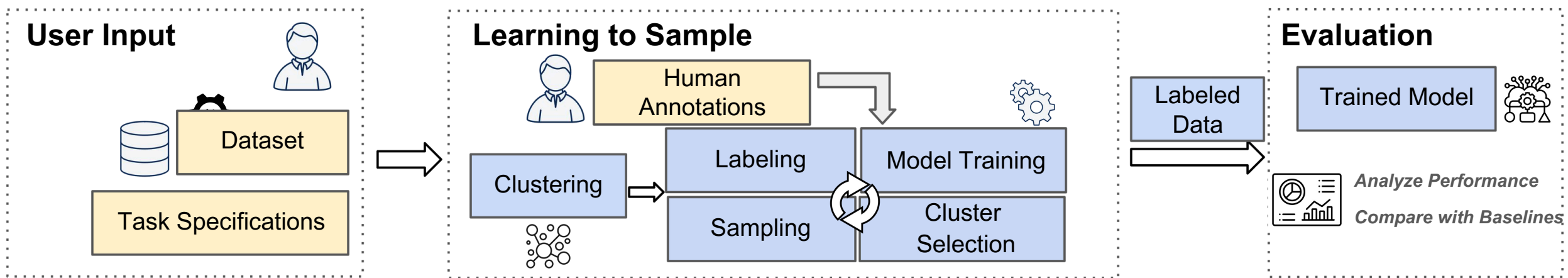


- LTS is a cost-effective approach to label data for training machine learning classifiers.
- It uses active learning and leverages Large Language Models (LLMs) to generate *pseudo-labels* for a selected subset of the data.
- By combining clustering with a 'multi-armed bandit' strategy, it selects a diverse and representative sample of ads.
- This strategy reduces the number of expensive LLM calls, making the process more efficient, and leads to robust and effective classifiers.

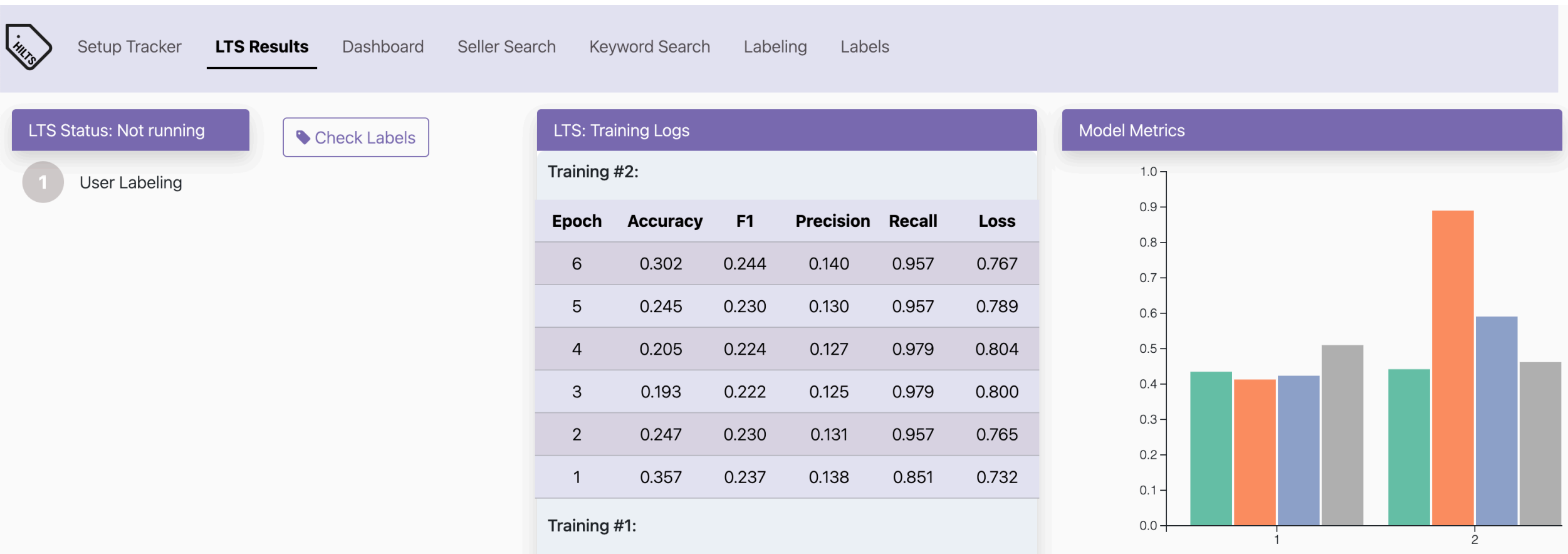
Challenge 3: LLMs are effective, but they can make mistakes

HILTS - Human in the Loop LTS

Getting high-quality labeled data from large, unbalanced datasets.



- HILTS supports human-AI collaboration: users actively participate in the labeling process by reviewing and correcting LLM-derived pseudo-labels.
- An easy-to-use visual interface allows users to define their triage criteria and incorporate domain knowledge to attain higher accuracy.
- Through an iterative process, users interact with HILTS to refine the training data, leading to improved model performance.



Results

Cost of LTS vs GPT-4: LTS incurs a one-time cost to create classifiers that can be subsequently used for multiple inferences; in contrast, we need to pay to classify each individual ad using GPT-4.

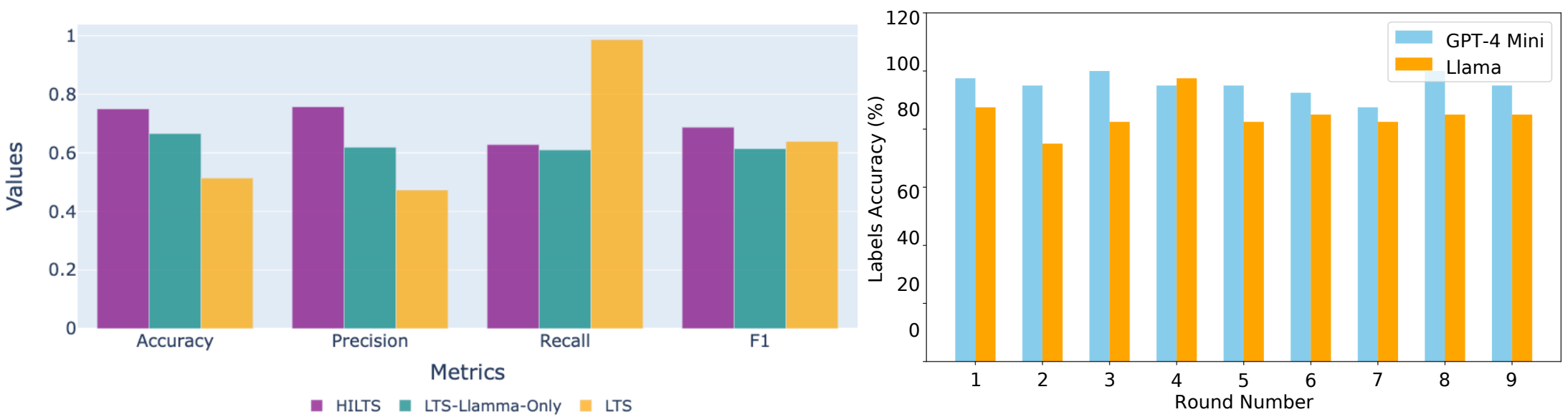
Task	Dataset Size	GPT-4 Cost	LTS Cost
Animal Products	700k	\$14,000	\$40
Small Leather Products	152k	\$3,350	\$40
Shark Products	15k	\$320	\$40

Performance Comparison (using F-1 measure) of the LTS model against zero-shot classification performed by different LLMs:

- The model performs substantially better than LLAMA and LLAVA-MM
- It performs comparably to GPT4, using only 6.2% of the total parameters and at a small fraction of the cost.

Task	LLAMA3 (8 Bi)	GPT4 (1.76 Tri)	LLAVA-MM (7 Bi)	LTS-text (110 Mi)
Animal Product	0.556	0.827	0.481	0.816
Leather Products	0.781	0.850	0.472	0.855
Shark Products	0.659	0.946	0.247	0.873

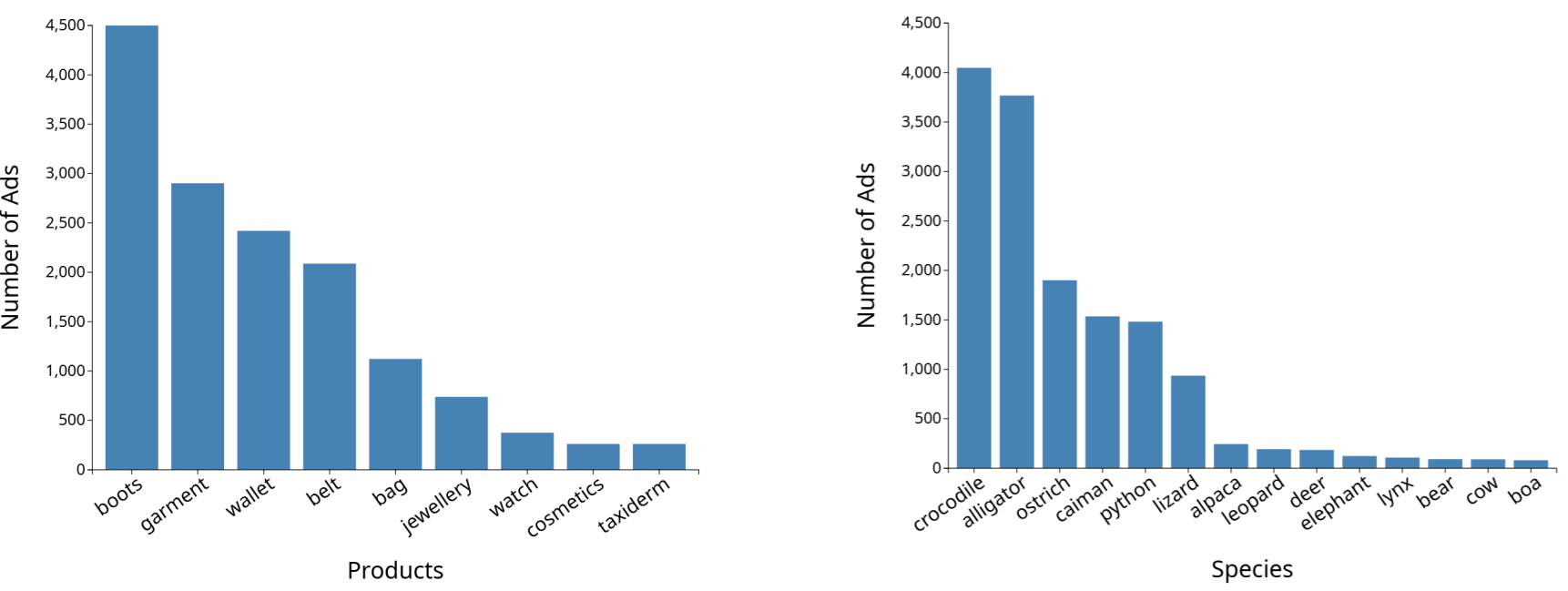
Getting high-quality labels using human in the loop to correct LLMs' mistakes leads to better models.



Research Findings

1. Small leather goods trade: identifying trends and platform vulnerabilities

- 67% (n=99) of sellers sell small leather products potentially made from protected species.
- Uncovered different types of sellers: specialized and opportunistic



2. The Online Trophy Trade of Sharks, Rays, and Chimaeras

- Despite policies on eBay, derivatives of restricted species (like CITES Appendix II sharks) are still traded.
- Identified high traffic in tiger shark jaws and parts from rare deep-sea species.

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